

Thesis Outline

Predicting Zoning Opposition

1. Introduction (3-5 pages)

- **1.1 Context:** Rise of zoning opposition and NIMBYism; protest risks for urban planning.
- **1.2 Case Study:** Austin's housing market, rapid growth, and history of zoning protests.
- **1.3 Research Objectives:** Forecasting per-parcel protest risks via DDPM and LogReg.

2. Literature Review (5-7 pages)

- **2.1 Zoning Opposition Dynamics:** NIMBYism, homevoter theory, neighborhood resistance.
- **2.2 Quantitative Urban Planning:** Predictive modeling applied to urban phenomena.
- **2.3 Generative Causal Models:** DDPM and Invariant Causal Prediction in tabular models.

3. Mixed-Methods Data and Methodology (8-10 pages) ✓

- ⊗ **3.1 Quantitative Sources & Panel:** Building the balanced property-year panel (282k parcels).
- ⊗ **3.2 Causal Prediction & Leakage:** Preventing temporal leakage across distinct training years.
- ⊗ **3.3 Qualitative Stakeholder Interviews:** Methodology and thematic coding of 10 interviews.

4. Predictive Modeling Approach (Quantitative) (5-8 pages) ✓

- ⊗ **4.1 Baseline Models:** Setting up Logistic Regression and XGBoost benchmarks.
- ⊗ **4.2 Conditional Diffusion (DDPM):** Training the generative model for tabular protest risk.
- ⊗ **4.3 Evaluation Metrics:** RMSE, MAE, CRPS, Scenario Std, out-of-sample validation.

5. Quantitative Results and Counterfactuals (8-10 pages) ✓

- ⊗ **5.1 Baseline Performance:** Feature importance and predictive accuracy of traditional models.
- ⊗ **5.2 Generative Models:** Evaluating DDPM accuracy and its realistic counterfactuals.
- ⊗ **5.3 Comparative Analysis & Policy:** Using evaluated models to inform proactive rezoning.

6. Qualitative Results: Stakeholder Perspectives (8-10 pages)

- **6.1 Planners and Advocates:** How institutional actors view algorithmic planning tools.
- **6.2 Neighborhood Leaders:** The tension between predictive efficiency and civic participation.
- **6.3 Democratic Implications:** Synthesizing the boundaries of acceptable use.

7. Synthesis and Conclusion (4-6 pages)

- **7.1 Integrating Findings:** Reconciling models with stakeholder demands for transparency.
- **7.2 Limitations:** ID crosswalk mismatch rates, assumption constraints, model bounding.
- **7.3 Future Work:** Scaling to other jurisdictions; NLP-parsed qualitative petition data.

Total Estimated Length: 41-56 pages (excluding Abstract, References, Appendices)